Foundations of the Geometric Vacuum Framework: A Synthesis of the Spectral Graph Theory Arc*

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The Geometric Vacuum (GeoVac) framework is a discretization scheme for non-relativistic quantum mechanics on natural geometries: the discrete graph Laplacian on a packed lattice of quantum-number labels replaces the continuous Laplace–Beltrami operator on the corresponding compact manifold. This synthesis paper assembles the framework's foundational arc—the packing construction, the discrete graph and its dynamics, the conformal equivalence with the unit S^3, the exchange constant taxonomy, the angular sparsity theorem, the Bargmann–Segal lattice, and the universal/Coulomb partition, together with the *reconvergence* node where these threads cohere (two-body selection rules from the tensor-product triple, the period classification, the Tannakian substrate and cosmic-Galois injection, and the forced/free seam)—into a single coherent narrative. The arc has a dependency structure worth naming: a single taproot (the packing construction, made physical by the Fock S^3 projection) fans out into the framework's machinery, then reconverges at the periods, Tannakian, and forced/free nodes, where the scattered structural findings cohere into one statement about what the framework forces and what it consumes. Four headline structural results frame the program: **[SYMBOLIC PROOF]** (a) the discrete Fock graph's continuum limit is conformally equivalent to the unit S^3 Laplace–Beltrami operator, verified by a suite of 18 symbolic checks (several limiting-case or definitional rather than logically independent, Paper 7); **[INTERNAL THEOREM]** (b) the angular sparsity of two-body matrix elements depends only on $l_{\max}$ and is bit-identical across five distinct central potentials, qualifying as a potential-independent theorem about spherical-fermion bases; **[PANEL-VERIFIED]** (c) the three-dimensional isotropic harmonic oscillator admits a discrete graph encoding on the Hardy space $H^2(S^5)$ that is bit-exactly π-free in exact rational arithmetic at every finite cutoff; **[OBSERVATION]** (d) the framework's results split cleanly along an A/D partition—an algebra-only universal sector that transfers free across central potentials, and a Dirac-operator-specific sector containing the spectrum, the calibration π, and the candidate fine-structure-constant combination of Paper 2 [29]. **[OBSERVATION]** A structural reading we make explicit in this synthesis is that the framework's substrate-level scale-freedom—the packing diameter d_0 cancelling out of every spectral quantity, and physical scales entering only through the Fock-projection focal length p_0—is the geometric expression of the same principle that conformal field theory embodies at the physics level (cf. the F-theorem of [28]). GeoVac provides a scale-free mathematical substrate; CFT is a scale-free physics theory living on such a substrate. This is not a claim that GeoVac is a CFT; it is a structural reflection consistent with the framework's reproduction of continuum CFT-on-sphere observables bit-exactly (documented in the operator-algebra arc, outside the foundations scope of this synthesis). We present GeoVac as a discretization framework that exploits natural geometry, not as a new foundation for quantum mechanics: the discrete graph and the continuous Laplace–Beltrami operator are dual descriptions connected by proven conformal maps, and the interpretive question of which description is more fundamental is left open. The framework's concrete computational advantages—$O(V)$ sparsity, integer eigenvalues on the unit S^3, angular momentum selection rules baked into the basis, bit-exactly rational matrix elements on the Bargmann–Segal lattice, and the universal sparsity bound—are independent of that interpretive question.

I. INTRODUCTION

The standard formulation of non-relativistic quantum mechanics presupposes a continuous configuration space and treats the Schrödinger operator $-\frac{1}{2}\nabla^2 + V(r)$ as the fundamental object. For the hydrogen atom in particular, Fock showed in 1935 [1] that the momentum-space Schrödinger equation maps by stereographic projection onto the free Laplacian on the unit three-sphere S^3. This observation has been developed extensively by Bargmann [2], Bander and Itzykson [3], Avery [8], Aquilanti and co-workers [9], and others working in the hyperspherical-momentum-space tradition. In each case, the S^3 structure is treated as an elegant reformulation of a known continuum problem: a useful change of variables for analytic work, not a foundation in its own right.

The Geometric Vacuum (GeoVac) framework [17, 18, 21] takes a complementary computational stance. Starting from a packing construction on an isotropic sphere, it builds a finite directed graph whose nodes are quantum-number labels (n, l, m) and whose edges are transition amplitudes generated by the $SU(2) \otimes SU(1,1)$ ladder algebra of hydrogen. The graph Laplacian on this lattice is dimensionless and scale-invariant; physical ener-

* Group 3 synthesis in the Geometric Vacuum series

gies emerge only upon imposing the Fock energy-shell constraint $p_0^2 = -2E$, which acts as a stereographic focal length. The framework's central reading, recorded in Paper 7 as an Observation (re-priced 2026-09-03), is that this discrete construction converges to the unit S^3 Laplace–Beltrami operator in the continuum limit; what is measured is the spectral-bound saturation and the s/p degeneracy recovery, not the operator convergence. The discrete and continuous descriptions are connected by a proven conformal equivalence, and we present them throughout as dual descriptions of the same physics; the interpretive question of ontological priority is explicitly left to the reader.

GeoVac differs from three nearby families of discretization schemes in ways worth naming at the outset. Lattice gauge theory discretizes the spacetime continuum on a hypercubic grid; GeoVac discretizes the quantum-number space and inherits the natural geometry of separable quantum systems. Tensor-network methods choose a basis adapted to entanglement structure; GeoVac chooses a basis adapted to angular momentum and the Fock conformal projection. Algebraic quantum field theory works at the level of operator algebras over Minkowski spacetime; GeoVac is non-relativistic and operates on compact Euclidean-signature manifolds. The rhetoric rule that governs the GeoVac papers (see the project's positioning statement) applies here: GeoVac is not a new foundation for quantum mechanics, and the synthesis presented in this paper does not assert ontological priority of either the discrete or the continuum description.

The Group 3 (foundations) arc comprises eleven papers, organized by a single dependency structure. *The taproot:* Paper 0 [17] states the packing axiom and derives the angular quantum numbers as packing indices; Paper 7 [21] makes it physical, verifying the conformal equivalence with the unit S^3 via a suite of 18 symbolic checks (several limiting-case or definitional) and extending to the multi-electron S^{3N-1} setting. *The fan-out* (the framework's working machinery): Paper 1 [18] sets up the spectral graph theory and reproduces the Rydberg quantum-number structure (the energies follow once the matched scale κ is applied); Paper 18 [22] classifies transcendental content via a six-tier exchange-constant taxonomy and the master Mellin engine; Paper 22 [24] states the potential-independent angular sparsity theorem; Paper 24 [25] constructs the Bargmann–Segal lattice and the seven-layer Coulomb/HO asymmetry; Paper 31 [26] states the $\mathcal{A}/D$ universal/Coulomb partition in spectral-triple language. *The reconvergence* (where the branches loop back to a unified foundational statement, §IX): Paper 54 [39] derives two-body selection rules from the tensor-product triple, Paper 55 [40] classifies every GeoVac period as cyclotomic mixed-Tate at level ≤ 4, Paper 56 [41] constructs the cosmic-Galois comparison map $U^*_{\mathrm{GV}} \to \mathcal{U}_4^{\mathrm{ab}} \rtimes SL_2$ (a homomorphism, not a closed immersion), and Paper 57 [42] catalogues the forced/free seam. An $O(V)$ time-evolution engine (Paper 6 [20], now an archived standalone tool) supplies the dynamical demonstration; this synthesis supersedes the earlier single-document synthesis (Paper 21 [23]).

Four headline structural results organize the rest of this synthesis. **(H1) [SYMBOLIC PROOF]** The conformal chain from the unit-S^3 Laplace–Beltrami operator to the Schrödinger equation is verified symbolically (Paper 7 [21]); **[OBSERVATION]** that the discrete Fock graph's continuum limit is that operator is a reading supported by numerical indications through $n_{\max} = 30$ (spectral-bound saturation, s/p-lift decay), not an established convergence (re-priced 2026-09-03), while **[INTERNAL THEOREM]** the state-space Gromov–Hausdorff convergence of the spinor spectral truncations is proved unconditionally with rate $(4/\pi)\log n_{\max}/n_{\max}$ (Paper 38 [36]). **(H2) [INTERNAL THEOREM]** The angular sparsity density of two-body matrix elements is potential-independent (Paper 22 [24]): a structural theorem of spherical-fermion bases, not a Coulomb-specific accident. **(H3) [PANEL-VERIFIED]** The 3D harmonic oscillator admits a discrete graph encoding on the Hardy sector of S^5 that is bit-exactly π-free (Paper 24 [25]). **(H4) [OBSERVATION]** The framework's results decompose cleanly into a universal algebra-only sector and a potential-specific Dirac-dependent sector (Paper 31 [26]). We discuss the fine-structure constant combination $K = \pi(B+F-\Delta)$ of Paper 2 [19] only in the open-questions section, in keeping with the project rule that the combination rule remains an *Observation* at the rule level despite three independent spectral homes for B, F, and Δ.

II. THE PACKING CONSTRUCTION AND THE GRAPH

GeoVac's discrete lattice is constructed by a packing argument that is prior to the Schrödinger equation [17]. Two axioms suffice.

Axiom 1 (binary distinguishability). A minimum of two distinguishable states is required to define a metric on an information-bearing manifold.

Axiom 2 (geometric indifference). In the absence of external constraints, the distribution of states on a homogeneous surface maximizes entropy and is therefore isotropic.

[SYMBOLIC PROOF] Applied to concentric isotropic shells in two dimensions, these axioms force the area-proportional state count on shell k to be $2k-1$. The shell capacities $\{1, 3, 5, 7, \ldots\}$ are isomorphic to the spherical harmonic basis Y_{lm} under the identification $l = k-1$ and $m = -l, \ldots, +l$. The angular quantum numbers (l, m) and the per-shell degeneracy $2l+1$ thus emerge from the packing geometry alone, without invoking the Schrödinger equation or the angular momentum operators $\hat{L}^2$, $\hat{L}_z$.

[OBSERVATION] The principal quantum number n enters as a cumulative grouping of the first n shells, producing the count $\sum_{k=1}^{n}(2k-1) = n^2$. Paper 0 is careful to flag that this grouping is natural but not uniquely forced by the two axioms. Similarly, the factor of two corresponding to spin multiplicity arises from the orientability of the compactified packing manifold (S^2), which the paper presents as a structural correspondence rather than a derivation. The strong claim is the angular sector (l, m); the other identifications are framed as structural correspondences.

From the packing to the graph

The packed lattice acquires a graph structure when the $SU(2) \otimes SU(1,1)$ ladder operators of the hydrogen dynamical algebra [4, 16] are imposed as edges:

$$L_{\pm}|n,l,m\rangle \propto |n,l,m\pm 1\rangle, \tag{1}$$

$$T_{\pm}|n,l,m\rangle \propto |n\pm 1,l,m\rangle. \tag{2}$$

The angular operators $L_{\pm}$ connect states within an n-shell along m; the radial operators $T_{\pm}$ connect states between adjacent n-shells at fixed l. **[PANEL-VERIFIED]** Paper 1 [18] demonstrates that the Casimir invariants built from the ladder operators reproduce the Rydberg quantum-number structure (n and $l(l+1)$) exactly; the energies $E_n = -1/(2n^2)$ follow only once the matched kinetic scale $\kappa = -1/16$ is applied, which the corpus records as an Observation (Paper 7). At the algebraic level the discrete and continuous descriptions therefore agree on the quantum-number skeleton, while the graph Laplacian's node-amplitude proxy for the s/p gap is large at finite cutoff (up to 37% at $n_{\max} = 5$ on the binary lattice, below 1% by $n_{\max} = 30$; the ℓ-blocks are disconnected, so this is a proxy and not a spectral gap — Paper 1 §III).

The universal kinetic constant $\kappa = -1/16$

[OBSERVATION] A second invariant of the packing geometry is the universal kinetic constant $\kappa = -1/16$. Its operational role is to map the dimensionless S^3 eigenvalues $\lambda_n = -(n^2-1)$ to the physical Rydberg spectrum via the energy-shell focal length $p_0 = Z/n$. Recent analysis [21, 22] has clarified its status: the value $\kappa = -1/16$ *coincides* with a geometric quantity computed independently of any energy calibration. The squared coupling between adjacent n-shells in the Gegenbauer eigenbasis is $c^2(n,l) = (1/4)[1-l(l+1)/(n(n+1))]$ (corrected 2026-09-03 from a printed 1/16 prefactor; the Chebyshev amplitude is 1/2), giving $c^2(n,0) = 1/4$ for $l = 0$; the quantity that coincides with 1/16 is the inverse Fock Jacobian $1/\Omega^4(0)$, equivalently $c^2(n,0)/4$. That the energy-calibration constant κ *equals* this geometric 1/16 is recorded as an *Observation*, not a derivation: the geometric value is rigid, but no mechanism is known that derives κ's calibration role from it (Paper 7 §III; Paper 18 places κ in the *conformal* tier of the exchange-constant taxonomy accordingly).

Status of the packing-versus-physics question

[OBSERVATION] The discrete lattice exists *a priori* from the packing construction, before any reference to the Coulomb potential or the Schrödinger equation. Its persistent match to known atomic structure—the Rydberg quantum-number structure, the chemical periodicity of multi-electron atoms (Paper 16 [31], outside the foundations group), nuclear magic numbers under a harmonic-oscillator basis (Paper 23 [32], applications group)—is therefore evidence that quantum-mechanical structure is consistent with a discrete underlying combinatorial topology, rather than a tautology. We present this as the *structural fit* between the packing construction and known atomic structure. The synthesis paper does not argue for the ontological priority of either description; the dual-description framing of Paper 7 governs the discussion.

III. THE S^3 CONFORMAL EQUIVALENCE

The theoretical foundation of GeoVac is the conformal equivalence between the discrete Fock graph and the unit S^3 Laplace–Beltrami operator established in Paper 7 [21]. This section states the result and explains its operational role.

The Fock projection and the chordal distance identity

[SYMBOLIC PROOF] Fock's 1935 construction [1] writes the momentum-space hydrogen Schrödinger equation

$$\left[\frac{p^2}{2} - E\right]\psi(\mathbf{p}) - \frac{1}{2\pi^2}\int \frac{\psi(\mathbf{p}')}{|\mathbf{p}-\mathbf{p}'|^2}\,\mathrm{d}^3p' = 0 \tag{3}$$

and applies the stereographic projection

$$u_i = \frac{2p_0\,p_i}{p^2+p_0^2}, \quad u_4 = \frac{p^2-p_0^2}{p^2+p_0^2}, \quad p_0 = \sqrt{-2E}, \tag{4}$$

which maps $\mathbf{p} \in \mathbb{R}^3$ onto the unit three-sphere $\mathbf{u} \in S^3 \subset \mathbb{R}^4$. The Coulomb kernel $1/|\mathbf{p}-\mathbf{p}'|^2$ becomes the inverse-square chordal distance on S^3, and the integral equation reduces to the eigenvalue problem

$$-\Delta_{S^3}\,\Psi_n = (n^2-1)\,\Psi_n, \tag{5}$$

free-particle motion on the unit sphere. The Coulomb potential is absorbed into the curvature of the sphere; the spectrum is determined by the topology of S^3 alone.

The 18 symbolic proofs

[SYMBOLIC PROOF] Paper 7 [21] verifies the conformal-equivalence chain through a suite of 18 symbolic checks implemented in `sympy` (Paper 7 flags several of them as limiting-case or definitional substitutions rather than logically independent theorems). The proofs cover the stereographic projection formulas, the unit sphere constraint, the chordal distance identity, the conformal factor identity, the conformal Laplacian transformation law, the integer eigenvalue spectrum, and the energy-shell projection from S^3 to $\mathbb{R}^3$. These tests live in `tests/test_fock_projection.py` and `tests/test_fock_laplacian.py` in the GeoVac repository and are run as topological integrity checks before any release. Their status as machine-verified algebraic identities pins the continuum-side conformal chain; the discrete-to-continuum step itself lies outside their scope (Paper 7 states this explicitly).

Schrödinger recovery in the continuum limit

[OBSERVATION] The graph Laplacian on the GeoVac lattice is read as converging, in the continuum limit $n_{\max} \to \infty$ with appropriate rescaling, to the round S^3 Laplace–Beltrami operator. This reading, recorded in Paper 7 with its numerical indications through $n_{\max} = 30$ (spectral-bound saturation, s/p-lift decay; no operator-convergence statement), remains unestablished for the graph Laplacian itself; what is now a theorem is the companion statement: **[INTERNAL THEOREM]** for the spinor (Dirac) spectral truncations it is an unconditional van Suijlekom state-space Gromov–Hausdorff convergence at rate $(4/\pi)\log n_{\max}/n_{\max}$ under the translation-seminorm metrization (Paper 38 [36], operator-algebra arc), with the $4/\pi$ constant universal across compact connected semisimple Lie groups (Paper 40 [37]: rigorous at rank 1, a rank-uniform analytical proof a named gap; a circle factor carries $2/\pi$); the scalar analogue of that theorem, for the Peter–Weyl spectral truncations of $L^2(S^3)$, is the prior-art state-space GH result of Gaudillot-Estrada–van Suijlekom [44]. Paper 7's 18 symbolic checks (several limiting-case or definitional) target the conformal-transformation chain, not a convergence statement, and the graph-side convergence itself has no theorem behind it: what the frozen test `tests/test_paper7_graph_convergence.py` pins is the saturation of the bipartite bound $\lambda_{\max} \to 2d_{\max} = 8$ and the block structure of the lattice Laplacian, not operator convergence (re-scoped 2026-09-03). Imposing the energy-shell constraint $p_0 = Z/n$ then maps the unit-sphere eigenvalues $\lambda_n = -(n^2-1)$ to the hydrogenic Rydberg series $E_n = -Z^2/(2n^2)$. The dimensionless graph carries no scale; the Rydberg energy is injected exclusively at the projection step. This separation between dimensionless topology and dimensional projection is the *dimensionless vacuum* viewpoint of Paper 7: the calibration content of physical observables is concentrated entirely in the focal-length parameter p_0.

Scale-freedom of the substrate and its reflection in conformal field theory

[OBSERVATION] The separation between dimensionless topology and dimensional projection described above is more than a computational convenience. It is the geometric expression of a principle that physics encounters in another guise: conformal field theory.

A conformal field theory is a quantum field theory whose action and correlators are invariant under rescaling of the underlying coordinates, together with the larger conformal group $SO(d, 2)$ in d spacetime dimensions. Operationally: a CFT has no preferred length scale, and its observables transform predictably under conformal transformations. The standard examples include free massless scalars and fermions on round S^d, where the partition functions $Z^{\mathrm{CFT}}_{S^d}$ are computable in closed form and reproduce universal coefficients (e.g. the F-theorem constant F_{S^d} [28]).

GeoVac's substrate exhibits an analogous property at the *geometric / mathematical* layer, prior to physics. The packing construction (Paper 0 [17]) takes a circle of diameter d_0—intended as "the Planck length" in spirit, but without committing to a specific physical value—and outputs a graph whose nodes carry the labels of the eigenspaces of the continuum S^3 operator (the pure rational eigenvalues n^2-1 belong to that continuum operator; the graph spectrum is not a level-by-level copy of it, Paper 7), the multiplicities are integers, the Wigner-symbol couplings are exact rationals, and the packing diameter d_0 *cancels out* of every spectral quantity. Shrink d_0 to the Planck length, grow it to a galactic radius, set it to anything in between: the substrate is the same. The eigenvalues that come out of the packing are atomic quantum numbers in interpretation, but they are not atomic-scale in construction. The packing produces a scale-free structural skeleton.

The Fock projection of Paper 7 [21] is a *conformal map*: stereographic projection from S^3 to $\mathbb{R}^3$ preserves angles but rescales lengths by a position-dependent conformal factor $\Omega(\mathbf{p}) = 2p_0/(p^2 + p_0^2)$. Physical scales enter the framework through exactly this factor, in the form of the focal-length parameter p_0. The scale-free substrate becomes scale-bearing physics only when a specific p_0 is supplied (e.g. $p_0 = Z/n$ for hydrogenic atomic levels). Different choices of p_0 correspond to different physical interpretations of the same combinatorial skeleton; the substrate is agnostic to the choice.

This is structurally the same principle that CFT scale-invariance embodies. A CFT is a physics theory with no preferred length scale at the level of its action and observables; GeoVac's graph is a mathematical object with no preferred length scale at the level of its eigenvalues and matrix elements. CFT scale-invariance is the physics side of the principle; the dimensionless graph is the substrate side. The two converge structurally because they realise the same property—scale-freedom—at different layers of description, and the Fock projection that connects them is the conformal map that implements the scale-free-to-scale-bearing transition.

This reading is not a claim that GeoVac *is* a CFT, nor that the discrete graph derives CFT scale-invariance. Those would overshoot: GeoVac provides a geometric / spectral substrate, and a CFT is a quantum field theory living on a substrate. But the framework's ability to reproduce continuum CFT-on-sphere partition functions bit-exactly (recently demonstrated for the conformally coupled scalar and Camporesi–Higuchi Dirac on S^3 and S^5 in the operator-algebra arc, outside the foundations scope of this synthesis but cross-referenced in the open-questions section below) is consistent with the substrate-side scale-freedom established here; no mechanism is claimed. The substrate is the right kind of mathematical object to host scale-free physics, which is a reading of the parallel, not an explanation of it.

Multi-electron extension to S^{3N-1}

Paper 7 [21] and Paper 21 [23] extend the construction to N-electron atoms by replacing S^3 with the unit sphere $S^{3N-1} \subset \mathbb{R}^{3N}$ under the $SO(3N)$ rotation group. The single-electron Fock projection becomes a multi-electron hyperspherical projection, and the N-fermion antisymmetry condition on S^{3N-1} reads as a representation-theoretic constraint on the angular sector. The chemical periodicity of the periodic table emerges from the S_N subgroup of $SO(3N)$ acting on the hyperspherical angular basis (see Paper 16, applications group, outside the foundations scope of this synthesis).

[SYMBOLIC PROOF] A particular consequence of the S^3 projection is the closed-form single-center two-electron repulsion integral $F^0(1\mathrm{s}, 1\mathrm{s}) = 5Z/8$, derived from the node-overlap formula for V_{ee} on S^3 in Paper 7 for s-orbital pairs. More generally, all single-center Slater F^k integrals reduce to exact rationals in arithmetic over $\mathbb{Q}$ (the hypergeometric Slater engine, `geovac/hypergeometric_slater.py`, which is not part of Paper 7's derivation). The single-center two-electron problem is thus closed in rational arithmetic; transcendentals appear only at the projection-to-energy step (see Section VII).

What Paper 7 does and does not claim

Paper 7 is careful to separate established mathematics from new contribution. The stereographic projection formulas, the unit sphere constraint, the chordal distance identity, the conformal Laplacian relation, and the integer eigenvalue spectrum are all known results from Fock, Bargmann, Bander–Itzykson, and Barut–Kleinert. The new contributions are: (a) the numerical indications (spectral-bound saturation and degeneracy recovery through $n_{\max} = 30$) that the discrete GeoVac graph converges to the continuous S^3 Laplace–Beltrami operator, recorded as an Observation; (b) the systematic symbolic verification of the conformal transformation chain; (c) the framing of physical scales as confined to the energy-shell projection; (d) the computational consequence that $O(V)$ sparse graph eigenvalue problems can replace continuous integration. Paper 7's explicit statement of the rhetoric—"the mathematics supports both the discrete-as-fundamental and the continuous-as-fundamental readings, and we leave the choice to the reader"—is the foundation of the dual-description framing carried through the rest of the foundations papers.

IV. ANGULAR SPARSITY AS A STRUCTURAL THEOREM

The angular sparsity theorem of Paper 22 [24] is the foundations-side statement of the structural mechanism behind GeoVac's exactly Q-linear composed Pauli-term count ($N_{\text{Pauli}} = 27.90 \times Q$ at fixed basis, Paper 14 [30]) at the qubit-encoding level. The theorem can be stated cleanly without reference to the Coulomb potential or to any specific radial wavefunction.

Theorem (Paper 22). [INTERNAL THEOREM] *For any N-fermion system whose single-particle orbitals factor as products of spherical harmonics and arbitrary radial functions, the fraction of nonzero two-body matrix elements is determined solely by angular momentum selection rules and is independent of the radial potential.*

The proof is a direct application of the Wigner–Eckart theorem and Condon–Shortley angular momentum algebra [13, 14]. The two-body matrix element of a rotationally invariant interaction factors into a sum of products of Gaunt coefficients (angular factor) and Slater F^k, G^k integrals (radial factor); the Gaunt coefficients vanish for quartets (l_a, l_b, l_c, l_d) that do not satisfy the $3j$-triangle inequality $|l_a - l_c| \le k \le l_a + l_c$ and the corresponding inequality for (l_b, l_d). Exact enumeration at fixed $l_{\max}$ yields the ERI densities reproduced in Table I.

Universality across potentials

[PANEL-VERIFIED] The bound has been verified numerically across five qualitatively distinct potentials—Coulomb $-Z/r$, isotropic harmonic oscillator $\frac{1}{2}m\omega^2 r^2$, Woods–Saxon nuclear single-particle potential, square well, and Yukawa $e^{-\mu r}/r$—and produces bit-identical sparsity patterns at matched orbital counts. The universality is structural: the radial-angular factorization of $1/r_{12}$ in spherical harmonics (used in quantum chemistry) and the Talmi–Moshinsky bracket factorization of

TABLE I. Pair-diagonal angular ERI density $D_{\mathrm{pd}}(l_{\max})$ from Paper 22 (per-pair $m_a = m_c$, $m_b = m_d$), which holds only for axially-symmetric or m-decoupled bases. *The composed pipeline realizes the physically-correct global-M_L Coulomb density $D(l_{\max})$, not this one* — $\sim 4\times$ larger (8.52% at $l_{\max} = 2$, 6.06% at $l_{\max} = 3$; Paper 22 Table). The "realizes" attribution was corrected 2026-08-30: the production enumerator returned D_{pd} only because of the wrong-sign-q coefficient bug fixed on 2026-08-29. Both are universal across Coulomb, harmonic oscillator, Woods–Saxon, square well, and Yukawa potentials.

$l_{\max}$	$D_{\mathrm{pd}}(l_{\max})$
0	100%
1	7.81%
2	2.76%
3	1.44%
4	0.90%
5	0.62%

central NN potentials in oscillator bases (used in nuclear shell models [15]) impose the same angular selection rules. GeoVac's contribution is to state the explicit quantitative density at each $l_{\max}$ and to frame it as a universal resource guarantee independent of the radial physics.

Spinor extension

[PANEL-VERIFIED] Paper 22's spinor extension shows that the angular sparsity bound survives, with mildly inflated prefactor, when the basis is lifted from Y_{lm} to the spinor spherical harmonics $\Omega_{\kappa m_j}$ labeled by the Dirac quantum numbers (κ, m_j). The density $d_{\mathrm{spinor}}(l_{\max})$ satisfies $d_{\mathrm{spinor}} \leq d_{\mathrm{scalar}}$ at every $l_{\max}$ tested, with the same triangle inequality controlling vanishing of the matrix elements at the spinor level. Tier-2 spin-orbit extensions of the framework therefore inherit the angular sparsity guarantee without modification.

Operational consequence

The angular sparsity theorem is what makes GeoVac's qubit-encoded Hamiltonians competitive for VQE/NISQ resource estimation. **[MEASURED]** The quantum-computing applications papers (Group 4) report Pauli-term counts exactly linear in the qubit count for the composed-geometry architecture across the molecular library ($N_{\mathrm{Pauli}} = 27.90 \times Q$ at fixed basis)—a $54\times$ to $317\times$ equal-qubit advantage in Pauli-term count over published Gaussian baselines (H_2O, $n_{\max} = 1$–3; the retired pair-diagonal rule gave $O(Q^{2.5})$ and $51\times$–$1712\times$). The structural reason for this advantage is the Gaunt-selection-rule sparsity quantified in Table I; the foundations-paper statement is the theorem itself, not the specific molecular benchmarks.

V. QUANTUM DYNAMICS ON THE GRAPH

Paper 6 [20] extends the framework from spectroscopy to time evolution. *Provenance note:* Paper 6 is archived within the series. Of the results quoted in this section, the unitarity benchmark and the resonant Rabi oscillation (item (b) below) were re-run and reproduce, with regression backing live in `tests/test_rabi_oscillation.py`; until 2026-09-01 that file was `tests/rabi_oscillation.py`, named so that collection skipped it, with its cases wired as a script rather than as tests. Items (a), (c) and (d) below are Paper 6's archived measurements with no live regression test (a named coverage gap in the claim matrix). The graph Hamiltonian H is sparse with $O(V)$ nonzero entries, so the Crank–Nicolson propagator

$$\psi(t + \Delta t) = \left(I + \tfrac{i}{2} H \Delta t\right)^{-1} \left(I - \tfrac{i}{2} H \Delta t\right) \psi(t) \qquad (6)$$

admits implementation at $O(V)$ cost per step using sparse matrix-vector products plus iterative linear solves. **[MEASURED]** Unitarity is preserved to 10^{-10} (measured 1.1×10^{-13}) over 10^3 time steps in the live hydrogen regression ($n_{\max} = 10$, 385 states); Paper 6's original H_2 benchmark reported machine precision over 10^4 steps (archived measurement).

[MEASURED] The applications demonstrated in Paper 6 are: (a) broadband molecular spectroscopy from a single delta-kick propagation, recovering 20 dipole-active electronic transitions of H_2 with 0.16% mean error in 33 seconds; (b) resonant Rabi oscillations matching the analytic Bloch–Siegert-corrected period to 0.41% with 99.98% coherent population transfer (live hydrogen run, $n_{\max} = 4$, 30 states); (c) ab initio molecular dynamics with nuclei coupled to graph forces via Velocity Verlet, conserving total energy to 0.0003%; (d) NVT-ensemble Langevin dynamics with stable thermal vibrations at 300 K. Only (b) and the unitarity benchmark carry live regression backing (see the provenance note above).

Paper 6 is not a foundational claim in the same sense as Papers 7, 18, 22, 24, or 31—it is an operational demonstration that the foundations support time evolution. Its role in the synthesis is to close the gap between the spectral statement of Papers 1 and 7 (the graph reproduces the Rydberg quantum-number structure) and the dynamical statement required for any quantum-chemistry application (the graph supports time evolution at the same $O(V)$ cost). Without Paper 6's dynamics engine, GeoVac would be limited to eigenvalue problems; with it, the framework extends to time-dependent quantum chemistry, spectroscopy, and molecular dynamics.

VI. THE BARGMANN–SEGAL LATTICE AND THE COULOMB/HO ASYMMETRY

Paper 24 [25] extends GeoVac to the three-dimensional isotropic harmonic oscillator (HO) and uncovers a structural asymmetry between the Coulomb and HO discretizations. The construction is built on the Bargmann–Segal transform [5, 6].

The discrete HO lattice on S^5

The Bargmann–Segal transform maps the 3D isotropic HO Hamiltonian to the Euler number operator $\hat{N}$ on the Fock–Bargmann space of holomorphic functions on $\mathbb{C}^3$. Restriction to the unit sphere $S^5 \subset \mathbb{C}^3$ identifies the eigenspaces with the symmetric $SU(3)$ irreducible representations $(N, 0)$, of dimension $(N+1)(N+2)/2$, which match the HO shell degeneracies exactly. The HO Hamiltonian becomes $\hbar\omega(\hat{N} + 3/2)$, with the eigenvalues linear in the shell index N.

The discrete graph encoding of this structure takes nodes (N, l, m_l) to be the labels of the $(N, 0)$-restricted hyperspherical harmonics on S^5, and edges to be dipole transitions $\Delta N = \pm 1$, $\Delta l = \pm 1$ at exact-rational squared matrix elements. **[PANEL-VERIFIED]** At $N_{\max} = 5$ the graph has 56 nodes and 165 edges, all matrix elements computed in exact `fractions.Fraction` arithmetic, with zero irrationals encountered. The single π in the entire Bargmann–Segal ecosystem is the Gaussian normalization π^{-3} of the continuous Bargmann measure, which the discrete graph never invokes.

The HO rigidity theorem

Theorem (Paper 24, HO rigidity). [CONDITIONAL] *The 3D isotropic harmonic oscillator is the unique central potential whose spectrum arises from the Euler number operator on the Hardy space $H^2(S^5)$ restricted to symmetric $SU(3)$ irreps $(N, 0)$.* Paper 24 states the named gap explicitly: the converse half—that SU(3) closure *forces* $V \propto r^2$—is an imported dynamical-symmetry result the paper flags as unverified at the pointer level.

This is the structural dual of the Fock rigidity theorem of Paper 23 (applications group), which states that the S^3 stereographic projection $p_0 = \sqrt{-2E_n}$ maps a one-electron central-field Hamiltonian onto the free Laplacian on S^3 if and only if the spectrum is l-independent within each n-shell, a property unique to $-Z/r$ by $SO(4)$ symmetry. Together the two rigidity theorems establish that the choice of central potential is sharply tied to the choice of natural manifold: $-Z/r$ to the second-order Riemannian Laplacian on S^3, $\frac{1}{2}m\omega^2 r^2$ to the first-order Euler operator on $H^2(S^5)$.

The Coulomb/HO asymmetry: a seven-layer stratification

Conventions. Sprint and track designations below are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository); "wall" is project shorthand for a structural obstruction established by multiple independent negative results.

[OBSERVATION] A central structural finding of Paper 24, sharpened progressively by the S^5 gauge-structure analysis (Paper 25 [33], QED/gauge group), a sprint on the wedge KMS state of the framework's Lorentzian extension, the spectral-action gravity arc (Paper 51 [38]), and the master Mellin engine, is that the Coulomb and HO discretizations differ along *seven* independent layers, identified in stages across 2025–2026. The asymmetry as currently catalogued reads:

1. **Spectrum-computing role of L_0.** On the Coulomb S^3 graph, the graph Laplacian L_0 is the spectrum-generating operator: the spectrum of $(D - A)$ approaches the continuum S^3 Laplace–Beltrami spectrum $n^2 - 1$ as $n_{\max} \to \infty$ (numerically, Paper 7), and the hydrogenic energies follow with the matched kinetic scale. On the Bargmann–Segal lattice, L_0 is the matter-propagator operator, not the Hamiltonian: the spectrum comes from the diagonal Euler operator $\hbar\omega(\hat{N}+3/2)$, and the graph adjacency encodes dipole transitions only.

2. **Calibration π. [PANEL-VERIFIED]** The Coulomb S^3 projection introduces calibration π as the Hopf-base measure (see Section VII); the HO Bargmann projection is bit-exactly π-free at every finite $N_{\max}$. This is one of the clearest concrete instances of an exchange constant being tied to a specific natural geometry.

3. **Wilson-physical-content.** The non-abelian Wilson gauge construction on $S^3 = SU(2)$ admits natural matter coupling (the gauge field acts on Pauli spinors carried by the graph nodes); the parallel SU(3) Wilson construction on the $(N, 0)$ Hardy tower fails the fixed-group-on-every-link requirement, because transitions between $(N, 0)$ and $(N + 1, 0)$ irreps are Clebsch–Gordan intertwiners rather than $SU(3)$ group elements. The natural gauge group of the Bargmann graph at the pure-gauge level is $U(1)$, not $SU(3)$.

4. **Modular-Hamiltonian structure of the wedge KMS state.** The Coulomb S^3 side admits a hemispheric-wedge KMS state with closed-form Hilbert–Schmidt orthogonality between the modular Hamiltonian and the wedge-restricted Dirac operator, with prefactor $1/\pi^2$ identifying as the Hopf-base measure signature; the HO Bargmann side does not admit the construction at all. This

is the operator-algebra-side reading of the asymmetry; for full details see the operator-algebra arc (Group 1).

5. **Spectral-action gravity termination. [SYMBOLIC PROOF]** The Camporesi–Higuchi Dirac spectral action on S^{2m+1} terminates at exactly $(m+1)$ power-law terms (Paper 51): only S^3 ($m = 1$) gives *pure Einstein gravity* (cosmological + Einstein–Hilbert, no R^2 or higher-curvature corrections), while S^5 ($m = 2$) terminates at three terms including an R^2 Gauss–Bonnet contribution. The mechanism is the Bernoulli identity for odd polynomials at half-integer shifts.

6. **Master Mellin engine chirality-parity. [PANEL-VERIFIED]** On the Camporesi–Higuchi S^3 triple at every finite $n_{\max}$, the Mellin source $\mathrm{Tr}(D^k e^{-tD^2})$ exhibits a bit-exact $\mathbb{Z}/2$ chirality-parity factorization (the plain trace vanishes at odd k, the supertrace at even k). On the Bargmann–Segal Hardy sector this fails on both structural conditions: the spectrum is strictly positive (no $\pm$-partner) and no half-integer-weight chirality grading exists on the $(N, 0)$ tower—propagating the asymmetry to the master-Mellin level.

7. *CM / Hodge-circle arithmetic of the compact flow.* On S^3 the compact KMS/Bisognano–Wichmann circle has $\pm$-symmetric integer generator spectrum, so its quarter-period point *is* a $\mathbb{Q}$-rational complex structure J, $J^2 = -1$, making the fundamental doublet a polarized weight-one $\mathbb{Q}$-Hodge structure with complex multiplication by $\mathbb{Q}(i)$ (Paper 56). The S^5 Hardy phase circle is equally compact and even carries a faithful μ_4 (the metaplectic zero-point $\frac{3}{2}$), yet admits no CM Hodge structure—its generator is strictly positive, the quarter point is $\mathbb{Q}$-rational only where it squares to $+1$, and the conjugate half a weight-one structure needs is exactly what the holomorphic-sector restriction discards. Conductor-four *period* content, by contrast, does transfer (Paper 24's companion non-layer).

The first three layers were established in Paper 24 and Paper 25 (QED/gauge group, S^5 gauge-structure paper); the fourth (wedge KMS) was identified in a 2026 sprint and folded into Paper 24's §V; the fifth (spectral-action gravity) comes from Paper 51; the sixth (master Mellin chirality-parity) from a June 2026 sprint; and the seventh (CM/Hodge-circle arithmetic) from the August 2026 seam arc. The first six layers stratify along the same operator-order divide: the Coulomb side uses a second-order Riemannian Laplacian with a nonlinear stereographic projection, and the HO side uses a first-order complex-analytic Euler operator with a linear-affine Bargmann projection.

What the asymmetry says about transcendentals

[OBSERVATION] The bit-exact π-freeness of the Bargmann–Segal lattice is a structurally clean statement about where transcendentals enter physical observables. π is not a universal feature of natural geometry discretizations; it is a specific structural cost of the S^3 stereographic projection. This refines the exchange-constant taxonomy of Paper 18 (see Section VII): calibration π is Coulomb-specific, tied to second-order Riemannian operators with nonlinear projections, and has no analog on the first-order HO side.

VII. THE EXCHANGE CONSTANTS TAXONOMY

[OBSERVATION] Paper 18 [22] catalogues where transcendental constants enter physical observables computed on the GeoVac graph. The taxonomy is a six-tier classification organized by the type of compactness mismatch each transcendental mediates between the discrete representation and the non-compact coordinate system used for measurement.

Six tiers

1. **Intrinsic.** Algebraic content native to the discrete graph: rational eigenvalues, integer multiplicities, exact-rational Slater F^k integrals on S^3. π-free in arithmetic over $\mathbb{Q}$.

2. **Conformal / calibration.** π entering through the Hopf measure of the stereographic projection; the universal kinetic constant $\kappa = -1/16$; the Hopf-base measure $\mathrm{Vol}(S^2)/4 = \pi$ identifying as the a_0^2 Seeley–DeWitt coefficient on the spinor bundle. This is the source of calibration π in the Coulomb sector.

3. **Embedding.** Transcendentals introduced by embedding the discrete graph into a continuum coordinate system that the graph does not natively know: the electron-electron cusp $1/r_{12}$ is the canonical example. The cusp is two-dimensional in the angular pair coordinate (α, θ_{12}) and cannot be reproduced by any single algebraic insertion; partial-wave Schwartz extrapolation $\sim -A/(l_{\max} + 2)^4$ is the standard treatment.

4. **Algebraic-implicit.** Quantities defined by a polynomial equation $P(R, \mu) = 0$ with coefficients in a known field extension, where pointwise diagonalization is a computational convenience rather than a mathematical necessity. The Level-3 adiabatic potential curves $\mu(R)$ on the hyperspherical fiber bundle live in $\mathbb{Q}(\pi, \sqrt{2})[R]$.

5. **Composition.** Single transcendental seeds appearing through algebraic reduction of integrals: the seed $e^a \cdot E_1(a)$ that closes the Level-2 prolate-spheroidal radial solver via partial fraction decomposition; the seed $\gamma(1, x)$ closing the Slater F^k Laguerre-moment decomposition.

6. **Inner-factor input data.** Parameter-tied transcendental content that the framework consumes as Layer-2 input rather than deriving from Layer-1 graph data. The Yukawa Dirichlet ring $\mathbb{Q}[y_1^{-2s}, \ldots, y_n^{-2s}]$ associated with the inner-factor of a Connes–Marcolli almost-commutative spectral triple is the canonical example; nuclear-structure scalars ($\langle r^2 \rangle_E$, Zemach radius r_Z, quadrupole moment Q_N) are also parameter-tied Layer-2 inputs. This tier captures the boundary between what GeoVac derives and what it consumes from external spectroscopy.

[OBSERVATION] The taxonomy is falsifiable: the claim is that the transcendental content of any GeoVac observable is determined entirely by the type of projection linking the compact discrete description to the non-compact coordinate system. No observable has been identified that violates the taxonomy; recent work on bound-state QED in the hydrogen-Lamb-shift sprint has confirmed that the framework-native one-loop contributions sit cleanly in the calibration and composition tiers, with multi-loop UV-renormalization counterterms moving the relevant content into the inner-factor input data tier.

The master Mellin engine

[CONDITIONAL] A structural sharpening of the taxonomy is the master Mellin engine identified in Paper 18 §III.7 and formalized in the operator-algebra arc of the project. The Weyl–Selberg identification of GeoVac's exchange constants with Mellin transforms of heat kernels

$$\mathcal{M}\left[\mathrm{Tr}\left(D^k \cdot e^{-tD^2}\right)\right](s) \tag{7}$$

indexes three sub-mechanisms by the integer operator order $k \in \{0, 1, 2\}$:

- **M1 ($k = 0$).** Trivial heat kernel collapses to the unit-radius sphere-volume quotient $2\,\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 4/\pi$ (numerically $\mathrm{Vol}(S^2)/\pi^2$; not the Hopf fibration's base-to-total ratio, corrected 2026-09-03). This is the signature of all calibration-π content in the Coulomb sector and appears as the leading constant in the GeoVac state-space Gromov–Hausdorff convergence rate.

- **M2 ($k = 2$).** Standard Seeley–DeWitt expansion of $\mathrm{Tr}\, e^{-tD^2}$. Produces $\sqrt{\pi}\,\mathbb{Q} \oplus \pi^2\,\mathbb{Q}$ at every integer s via Jacobi θ-inversion on the half-integer-shifted Dirac spectrum.

- **M3 ($k = 1$).** Mellin transform of $\mathrm{Tr}(D \cdot e^{-tD^2})$. Produces the quarter-integer Hurwitz / vertex-parity content (Catalan's G, Dirichlet $\beta(2k)$) that appears in two-loop QED vertex corrections on S^3.

The master Mellin engine reading is that M1, M2, M3 are not three independent mechanisms but three sub-cases of the same Mellin transform on the GeoVac spectral triple, indexed by which power of the Dirac operator appears in the trace. Each sub-mechanism has a characteristic domain of observable: M1 governs state-space Gromov–Hausdorff convergence rates and Hopf-measure normalizations, M2 governs heat-kernel and spectral-action coefficients, and M3 governs vertex-restricted parity-character observables. Mixing the operator order of the question with the operator order of the answer produces the structural degeneracies catalogued in the project's sprint record.

The Layer-1 / Layer-2 split

[OBSERVATION] A particularly clean operational consequence of the taxonomy is the two-layer architecture made explicit by Paper 34 (catalogue paper, applications group): the bare combinatorial graph (Layer 1, π-free in $\mathbb{Q}[\sqrt{d}]$) versus a finite family of named projections (Layer 2) that add specific variables, dimensions, and transcendentals when the graph is mapped to a physical observable. Paper 34 documents 28 projections to date; each appears in the taxonomy of Paper 18 as either an intrinsic, calibration, embedding, algebraic-implicit, composition, or inner-factor input data mechanism. The two papers are complementary: Paper 18 organizes by type of compactness mismatch (what *kind* of transcendental appears), and Paper 34 organizes by named mechanism (*which* projection appears in *which* observable).

VIII. THE UNIVERSAL/COULOMB PARTITION

[OBSERVATION] Paper 31 [26] states the cleanest single articulation of what GeoVac claims as universal versus what it claims as potential-tied. The framework is read as a discrete spectral triple $(\mathcal{A}, \mathcal{H}, D)$ in the sense of Connes [10] and the graph-spectral-triple lineage of Marcolli and van Suijlekom [11, 12]: $\mathcal{A}$ is an algebra of functions on the Fock-projected discrete graph, $\mathcal{H}$ is a Hilbert space of scalar and spinor states, and D is a Dirac operator inherited from the Camporesi–Higuchi spectrum on S^3 [7].

The A/D split

The structural decomposition reads:

Universal sector ($\mathcal{A}$-only). Results that depend only on the algebra $\mathcal{A}$ and the angular structure of the basis, and *not* on the specific choice of Dirac operator. These transfer free across central potentials. Examples:

- Angular sparsity (Paper 22): ERI density $D(l_{\max})$ depends only on the spherical-harmonic angular factorization.
- Composed scaling (Paper 14, applications group): the exactly Q-linear Pauli-term count of composed-geometry qubit encodings ($N_{\text{Pauli}} = 27.90 \times Q$ at fixed basis) depends on the Gaunt selection rules.
- Selection rules: $\Delta l = \pm 1$, $\Delta m = 0, \pm 1$ for dipole transitions; the $3j$-triangle inequality at every vertex of the two-body integral.

Potential-specific sector (D-dependent). Results that depend on the specific spectrum and matrix elements of the Dirac operator D, and therefore on the specific central potential. These require per-system rederivation. Examples:

- The spectrum itself: $n^2 - 1$ for Coulomb, $N + 3/2$ for HO.
- The calibration $\kappa = -1/16$ (Coulomb-specific; no analog on the Bargmann lattice).
- Transcendental content: π for Coulomb via the Hopf measure, π-free for HO.
- Rigidity theorems: Fock rigidity (Paper 23) pins the Coulomb geometry to S^3; HO rigidity (Paper 24) pins the HO geometry to $H^2(S^5)$ with $(N, 0)$ symmetric irreps.
- The candidate fine-structure-constant combination $K = \pi(B + F - \Delta)$ of Paper 2: lives entirely in the Coulomb-specific sector by its dependence on the Camporesi–Higuchi Dirac spectrum on S^3.

Why the partition matters operationally

The partition organizes Papers 22, 23, 24, and 25 into a single conceptual structure. Paper 22's angular sparsity theorem lives in the universal sector and transfers free between hydrogen and the deuteron; Paper 23's nuclear shell model qubit Hamiltonians inherit the angular sparsity automatically but require a fresh per-potential analysis of the Dirac sector. Paper 24's HO rigidity and the π-free certificate live in the potential-specific sector and explicitly do *not* transfer to the Coulomb side—the Bargmann-style first-order operator has no analog on S^3. Paper 25 (group 5, QED/gauge) on the SU(2) Wilson lattice gauge structure of the Hopf graph also lives in the potential-specific sector: it depends on the parallelizability of $S^3 = SU(2)$, which has no analog on S^5.

The operational utility of the partition is that anyone porting GeoVac to a new system can read off, from the universal sector, what structural properties they inherit for free, and from the potential-specific sector, what calibrations they need to derive afresh. For nuclear shell models, gauge-theoretic extensions, and oscillator-based quantum dots, the universal sector provides angular sparsity and composed scaling automatically; the potential-specific sector has to be redone for each system.

Status as a spectral-triple-language reformulation

Paper 31 is careful to frame the partition as a structural reformulation, not a new theorem. The angular sparsity theorem (Paper 22), the Fock rigidity theorem (Paper 23), and the HO rigidity theorem (Paper 24) are each established results; the partition is the observation that they decompose cleanly along the algebra-only / Dirac-dependent divide. The framing uses the language of NCG and spectral triples because that language has the structural vocabulary to articulate the divide; the underlying physics content is unchanged.

IX. THE RECONVERGENCE: TWO-BODY STRUCTURE, PERIODS, AND THE TANNAKIAN SUBSTRATE

The foundations arc described so far *fans out* from a single taproot. The packing construction (Paper 0), made physical by the Fock S^3 projection (Paper 7), is the root; from it the framework branches into spectral graph methods (Paper 1), the exchange-constant taxonomy (Paper 18), angular sparsity (Paper 22), the Bargmann–Segal lattice (Paper 24), and the $\mathcal{A}/D$ partition (Paper 31). Four more recent papers do the opposite—they *reconverge*, looping the scattered structural findings back into a single foundational statement about what the framework autonomously generates and what it consumes.

A. Two-body structure (Paper 54)

The $\mathcal{A}/D$ partition of Paper 31 is, in the single-particle setting, a sorting of results into algebra-only and Dirac-dependent. Paper 54 tests it at the two-body level by constructing the tensor-product spectral triple $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^3}$ and asking whether the gauged spectral action generates the inter-particle Coulomb interaction from the framework's own structure. The free tensor-product Dirac $D_{\text{total}} = D_1 \otimes I + \gamma_1 \otimes D_2$ satisfies $\{D, \gamma\} = 0$ exactly, so the free spectral action factorizes and contains no interaction—the correct statement that free particles do not interact. **[MEASURED]** Gauging via the inner fluctuation of the truncated operator-system algebra produces a genuine, non-factorizable two-body interaction (connected fraction 77% at $n_{\max} = 2$, falling to 32% at $n_{\max} = 3$— cutoff-sensitive, like the radial Pearson coefficient below). **[PANEL-VERIFIED]** Its angular structure is exactly what the universal sector predicts: 100% m-conserving, 100% Gaunt-compatible, pure $k = 0$ monopole at both tested cutoffs. Its radial weights, however, are *not* proportional to the Coulomb $1/r_{12}$ kernel (Pearson $r = 0.41$–0.58, *decreasing* with $n_{\max}$—structural, not finite-size). This delineates the partition at the two-body level: the algebra $\mathcal{A}$ determines *which* channels couple (angular selection rules, universal across central potentials by Paper 22), and the Dirac operator / metric determines *how strongly* (radial coupling strengths, potential-specific). The recurring "multi-focal composition wall" sits exactly on this $\mathcal{A}/D$ boundary, and the reason is mathematical: the spectral action computes Seeley–DeWitt heat-kernel coefficients (metric curvature), while $1/r_{12}$ is a Green's-function kernel; the Coulomb potential enters through the Fock projection's chordal-distance identity (Paper 7), not through the spectral action.

B. Periods (Paper 55)

The master Mellin engine of Paper 18 indexes the framework's π-content by operator order $k \in \{0, 1, 2\}$ in $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$. **[INTERNAL THEOREM]** Paper 55 completes this engine into a period-theoretic statement: every transcendental that has appeared in a GeoVac spectral observable—integer powers of π, odd zeta values $\zeta(2k+1)$, Catalan $G = \beta(2)$ and Dirichlet-β values, and a depth-k loop tower—sits in cyclotomic mixed-Tate periods at level ≤ 4 over $\mathbb{Z}[i, 1/2]$, stratified by Mellin slot: $M_1 \subset \mathbb{Q}[\pi, \pi^{-1}]$ (pure-Tate, depth 0), $M_2 \subset \bigoplus_k \pi^{2k}\mathbb{Q}$ (pure-Tate even-weight, depth 0, a strict refinement of the Fathizadeh–Marcolli classification [45] specialized to the static S^3 sub-case), and $M_3 \subset \mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4 (cyclotomic mixed-Tate, via Hurwitz zeta at quarter-integer shifts and the Deligne–Glanois descent [46]; the vertex-parity-as-χ_{-4} correspondence of Paper 28 [34]). This is the period-theoretic completion of the $\mathcal{A}/D$ partition: the universal sector's transcendentals are pinned to named cyclotomic mixed-Tate rings rather than chased indefinitely. (Discipline check folded in here: the constant $S_{\min}$, long carried as "PSLQ-irreducible," is now resolved as $S_{\min} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ in closed form—the prior label was a basis-coverage artifact.)

C. The Tannakian substrate and the cosmic-Galois injection (Paper 56)

If the framework's periods all live in a single named period ring, the natural categorical home is a motivic Galois group. **[PANEL-VERIFIED]** Paper 56 assembles, bit-exact on rational panels (thousands of zero residuals across $n_{\max} \in \{2, 3, 4\}$), the Tannakian substrate of the GeoVac spectral triple at finite cutoff: a cofiltered pro-system of finite-cutoff Hopf-algebra truncations, an abelian symmetric-monoidal rigid representation category, a faithful exact $\otimes$-fiber functor, and an explicit inclusion of the candidate cosmic-Galois group $U^*_{\mathrm{Levi}} = \mathbb{G}_a^{3N} \rtimes SL_2$ into $\mathrm{Aut}^{\otimes}(\omega)$. **[INTERNAL THEOREM]** The load-bearing result is the *cosmic-Galois injection*: a homomorphism (not a closed immersion — the M3 column is rank 2, its parity-blind per-sector restriction rank 1, both $\ll N$, under Reading A; Paper 56) $U^*_{\mathrm{GV}} \to \mathcal{U}_4^{\mathrm{ab}} \rtimes SL_2$ of the framework's finite-cutoff Galois group into the abelianization of the unipotent radical of the level-4 motivic Galois group $\mathcal{G}_4 = \mathcal{G}_{\mathrm{MT}(\mathbb{Z}[i,1/2])}$ (Deligne 2010, Glanois 2015). The comparison target is the *level-4* group, not Brown's $\mathcal{G}_{\mathrm{MT}(\mathbb{Z})}$: Eskandari–Murty–Nemoto 2025 place Catalan G at level 4 (mixed Tate over $\mathbb{Q}(i)$). That G is *not* a period of $\mathrm{MT}(\mathbb{Z})$ is expected but not proven, so this motivates the refinement rather than forcing it. Two scope qualifiers are essential and stated honestly: (i) GeoVac *realizes the abelianization* of the level-4 cosmic Galois group (Reading A)—its primitive Hopf substrate is cocommutative, confirmed bit-exact at the JLO entire-cyclic-cocycle level at depth 2, so the image lands in $\mathcal{U}_4^{\mathrm{ab}} \cong \mathbb{G}_a^{\infty}$ rather than the full free non-abelian unipotent; (ii) *infinite-cutoff equality is not claimed*—the injection direction is closed at theorem grade at finite cutoff, while the converse equality on depth-≥ 2 content is the named deep forward question.

The elliptic layer (Paper 59 [43]). The injection above lands in the *mixed-Tate* (genus-zero) cosmic Galois; the elliptic (genus-one) layer that Paper 56 reaches for—a Hain–Brown mixed-elliptic-motive target [47, 48]—came back negative on the abstract Sym^2 panel, structurally because GeoVac's periods are $\mathbb{Q}(i)$-CM-pure rather than generically modular. **[MEASURED]** Paper 59 supplies a concrete elliptic period from the chemistry side: the three-center electron-repulsion integral, read in momentum space, is a two-scale Bessel moment on the Legendre / $\Gamma(2)$ family. The load-bearing arithmetic is *fibre-independent*: the fibre period is $K = \frac{\pi}{2}\theta_3^2$ and θ_3^2 is the theta series of $\mathbb{Z}[i]$, so the family carries conductor-4 arithmetic at every fibre—the same $\mathbb{Q}(i)$ the Dirac Kramers doublet of Paper 56 realizes, by a *different* forcing (level rather than spin): the two routes are convergent, not common-cause, and no comparison map between them has been exhibited. The physical density exchange acts on the modulus as $\rho \mapsto 1/\rho$, so the observable descends to $X_0(2) = X(2)/\langle\iota\rangle$; its physically distinguished fibre is the coincident-scale *pure-Tate cusp* $\rho = 1$, while the $\tau = i$ CM fibre is a visited member of the exchange orbit $\{\frac{1}{2}, 2\}$, not a pinned one. The integrated observable is an Eisenstein / complex-multiplication *Bessel moment*—an irregular (exponential) period, *not* a cusp-form L-value, of which a $\Gamma(2)$ multiple modular value is only the regular $D \to 0$ shadow. Its value is now certified to 66 digits, and the guarded searches are decisive: no low-height classical combination through weight three, including the Catalan-completed ring at height ≤ 10—so no measured conductor-4 content exists in the physical value at reachable heights (the arithmetic lives in the family and the shadow), and its explicit finite realisation remains open. Two honest bounds keep this in proportion. First, the level is $\Gamma(2)$, the classical *universal* Legendre level (below the two-loop sunrise's $\Gamma_1(6)$): landing on the Legendre family is expected of any four-branch-point curve, so the informative content is the rational modulus map $\lambda = 1 - \rho$, the $X_0(2)$ descent, and the physical Bessel twist's level promotion $\Gamma(2) \to \Gamma(4)$, not an exotic level. Second, whether the observable reaches the *generic* (non-CM, full-SL_2, $E_4/E_6/\Delta$) Hain–Brown locus is open. **[SYMBOLIC + MEASURED]** What the irreducibility *does* certify is structural: because the rank-four connection is the Fourier–Laplace transform of an irreducible rank-one connection—hence simple, with no invariant *coordinate* subspace in its monodromy frame—no closed form arises through any factorization, so the period is not reducible to the mixed-Tate layer by factorization (the genus-one label itself is set by the curve's two distinct Fock scales). On the physical contour the two layers demonstrably touch at the pure-Tate diagonal cusp; the seam's two upgrade tests—a natural comparison map, and a measured $\beta(2)$ in the physical value—are respectively tautology-blocked and answered negative at clean heights. The elliptic layer is thus a concrete, irreducible genus-one *instance* that corroborates rather than lifts the Hain–Brown negative—not filled to the generic modular locus.

D. The forced/free seam (Paper 57)

[OBSERVATION] The reconvergence closes with an observations paper that consolidates, across the corpus, the boundary between what the framework forces and what it consumes. Paper 57 catalogues the forced entries (graph spectrum, the S^3 outer manifold, the gauge group $U(1) \times SU(2) \times SU(3)$ from Bertrand $\times$ Hopf-tower truncation, the master Mellin engine, two-term-exact gravity), the single admitted-not-forced entry (the quaternionic factor $\mathbb{H}$), and the calibration entries (Yukawa values, generation count, cutoff moments, atomic screening exponents, multi-loop QED counterterms). It identifies the cross-domain unity pattern—the same multi-focal composition wall recurs in six independent settings—and tests a packing-reachability discriminator (an entry is forced iff a witness derivation exists from the packing axiom plus the real-spectral-triple axioms plus Hopf-tower truncation plus Bertrand). The packing-reachable tag is the F/C label by construction, so its 98.3% agreement across the 60-entry catalogue is an internal-consistency check, not an independent discriminator (Paper 57 §6.1); the genuine discriminator signal is the two-family decomposition below. The free side splits into two structurally distinct families: a multi-focal composition family (the framework can build it but cannot autonomously pick a number) and an inner-factor input-data family (categorical to the outer-factor mechanism; lives elsewhere by structure). This is the foundational reconvergence: the period classification (Paper 55) and the Tannakian injection (Paper 56) place theorem-grade boundaries around the empirical forced/free observation—with the $\mathcal{A}/D$ partition (Paper 31) supplying the spectral-triple language rather than a further theorem— though Paper 57 holds the forcing-iff-property principle itself open as a preliminary observation (it is an observations paper, not a theorem paper). The framework's identity is precisely this seam—it maps the structural skeleton of quantum mechanics (the compact, Peter–Weyl, discrete regime) and consumes, rather than generates, the continuum calibration data the skeleton does not reach.

X. OPEN QUESTIONS AND OUTLOOK

The foundations arc as catalogued here leaves five named structural questions open.

The α combination rule

Paper 2 [19] (observations group) presents the candidate combination

$$K = \pi(B + F - \Delta), \qquad K \stackrel{?}{=} \alpha^{-1}, \tag{8}$$

where $B = 42$ is a finite Casimir trace at the truncation $m = 3$, $F = \zeta_R(2) = \pi^2/6$ is the Fock-degeneracy Dirichlet series at the packing exponent $d_{\max} = 4$, and $\Delta = 1/40$ is the inverse Dirac mode count $g_3^{\mathrm{Dirac}}(S^3) = 2(n+1)(n+2)|_{n=3}$. Each ingredient now has a structurally identified spectral home; twelve mechanisms have been eliminated as candidate single-principle derivations of the sum [22]. The combination rule itself remains an *Observation* at the rule level (never a conjecture or derivation, per the project's hard rule): there is no first-principles derivation explaining why the additive combination of three structurally independent objects equals α^{-1} at 8.8×10^{-8}. The synthesis treats this as an open structural question and does not assert priority for the observed identification.

The second packing axiom (W3)

[OPEN] The packing construction of Paper 0 generates the angular sector (l, m) and provides natural correspondences for n and spin multiplicity, but it does not produce calibration data such as Yukawa couplings or the number of fermion generations. These appear in the exchange-constant taxonomy as Layer-2 inner-factor input data: the framework consumes them rather than deriving them. The open question labeled W3 in the project's internal taxonomy is whether a second packing-axiom-like construction generates calibration data from a similarly minimal principle. A 2026 sprint surveying candidates in the spectral-zeta / master-Mellin-engine ring returned uniformly negative results across PMNS angles, lepton masses, and a $\sim 21{,}000$-form mechanical basis [27]. The second-packing-axiom question is open and is structurally distinct from "find a packing on a different geometry"—the latter has been answered for the HO via the Bargmann–Segal lattice without addressing W3.

Cross-manifold extensions

The Coulomb and HO discretizations live on S^3 and the Hardy sector of S^5 respectively. **[OPEN]** A natural question is whether a coupled system (electronic plus nuclear, or two distinct potential regimes within one molecule) admits a tensor-product spectral triple of the form $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$. Within the foundations arc, Paper 24's seven-layer Coulomb/HO asymmetry documents structural obstructions to such a construction at the NCG-framework level. The operator-algebra arc (Group 1) has formally named this as a frontier-of-field open problem; no published framework currently supports the cross-manifold tensor product.

The multi-focal composition wall

[OPEN] The framework's composition rules for multiple Fock-style projections constitute an active research direction. The internal taxonomy labels three sub-walls: W1 (within-framework architectural gaps: cross-register coordinate operators, magnetization-distribution operators, frozen-core cross-center screening), W2 (multi-loop UV/IR composition and cross-manifold extensions), and W3 (inner-factor calibration data, discussed above). W1a/b/c have been closed at the first-pass level by the Phase C work documented in the applications group; W2a (multi-loop renormalization) remains open at the field-theory frontier of the operator-algebra arc. None of these walls falsify the foundations as presented in this synthesis; they sharpen the open-question boundary of the framework.

The elliptic layer of the cosmic Galois

[OPEN] Paper 56's cosmic-Galois injection lands in the mixed-Tate (genus-zero) group $\mathcal{G}_4$; the elliptic (genus-one) layer it reaches for—the Hain–Brown mixed-elliptic-motive target over $M_{1,1}$—came back negative, because GeoVac's periods are $\mathbb{Q}(i)$-CM-pure rather than generically modular. Paper 59 now instantiates that layer with a concrete, irreducible genus-one period (the three-center integral's elliptic Bessel moment on the Legendre/$\Gamma(2)$ family, descending to $X_0(2)$), carrying the same $\mathbb{Q}(i)$ arithmetic at the *family* level (θ_3^2, the $\mathbb{Z}[i]$ theta series) by level-forcing rather than spin-forcing—convergent with the Dirac route, not common-cause. The open question is whether any GeoVac observable reaches the *generic* (non-CM, full-SL_2) Hain–Brown locus—a lift that would unify the genus-zero and genus-one layers (the Hain–Brown mixed-elliptic completion). On the physical contour the two layers demonstrably meet at the pure-Tate cusp; no natural comparison map has been exhibited, and the measured seam test ($\beta(2)$ in the physical value) is negative at clean heights—a lift beyond convergence is unestablished.

XI. CONCLUSION

The Group 3 foundations arc of GeoVac establishes a small set of structural results that together provide a coherent discretization of non-relativistic quantum mechanics on natural geometries. **[SYMBOLIC PROOF]** The conformal chain from the unit-S^3 Laplace–Beltrami operator onward is verified symbolically (Paper 7 [21]); **[OBSERVATION]** the discrete Fock graph's convergence to that operator is a reading with numerical indications, not an established result, while the state-space GH convergence of the spinor spectral truncations is proved with explicit rate in Paper 38 [36]; **[INTERNAL THEOREM]** the angular sparsity of two-body matrix elements is potential-independent (Paper 22 [24]); the 3D harmonic oscillator admits a bit-exactly π-free discrete encoding on $H^2(S^5)$ (Paper 24 [25]); the framework's results decompose cleanly into an algebra-only universal sector and a Dirac-dependent potential-specific sector (Paper 31 [26]); the exchange-constant taxonomy of Paper 18 [22] catalogues where transcendentals enter physical observables and identifies a **[CONDITIONAL]** master Mellin engine at three operator orders as the generator of the π-content of the projection chains case-checked so far (Paper 32 [35]'s case-exhaustion theorem, over Paper 34 §III.1–15, with its (i)$\Leftrightarrow$(ii) leg resting on empirical verification); the packing axiom of Paper 0 [17] generates the angular sector geometrically; the spectral graph theory of Paper 1 [18] and the $O(V)$ time evolution of Paper 6 [20] provide the computational machinery.

The framework's concrete advantages are independent of the interpretive question of which description—discrete or continuous—is more fundamental. $O(V)$ sparsity, integer eigenvalues on the unit S^3, angular momentum selection rules baked into the basis, bit-exactly rational matrix elements on the Bargmann–Segal lattice, and the universal sparsity bound at fixed $l_{\max}$ are statements about the discrete graph; their continuum counterparts are theorems about the Laplace–Beltrami operator or the Euler operator on the corresponding natural manifold; the conformal equivalence guarantees consistency. The discrete and continuous readings are dual descriptions connected by proven maps, and the interpretive question is left open.

The open questions catalogued in Section X are substantive but do not destabilize the structural results of the foundations arc. The α combination rule remains an Observation; the second-packing-axiom question is open; cross-manifold extensions are blocked at the NCG-framework level; the multi-focal composition wall is an active research frontier; and the elliptic (genus-one) layer of the cosmic Galois is instantiated but not lifted—the generic Hain–Brown locus stays open. Each of these is a frontier-of-field question rather than a foundational concern, and the synthesis treats them as such.

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